

Project Initialization and Planning Phase

Date	4 October 2026
Team ID	N/A
Project Title	Student Mental Health Analysis (Tableau Dashboards & Story)
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To analyse student mental health data and present the findings through interactive Tableau dashboards and a captioned story, so that colleges can see which factors relate to student mental health risk and plan support.
Scope	One dataset of 1,000 students with 16 fields (gender, stress, depression, heart rate variability, sleep, study hours, academic score, grades and mental health risk). Delivered as Tableau Public dashboards, a story and a Flask ePortfolio site. It does not cover clinical diagnosis, prediction models or live data collection, and the dataset appears simulated, so findings are illustrative.
Problem Statement	
Description	Student mental health data sits in a long spreadsheet that is hard to read, so counsellors and mentors cannot easily see how gender, study habits, sleep and stress relate to mental health risk and end up relying on assumptions.
Impact	Clear visuals let counsellors and mentors test their assumptions against evidence. Here, stress, depression and risk are almost the same across genders (high risk 23-25%) and study hours do not predict academic score ($R^2 \approx 0$), so broad, inclusive support is a better fit than targeting one group.
Proposed Solution	

Approach	Load and check the dataset in Tableau, build one chart per question (gender comparisons, risk mix, study hours vs academic score, grades), combine them into a filterable dashboard with KPI tiles, add a captioned story that ends with a takeaway, then embed the Tableau Public views in a Flask ePortfolio site.
Key Features	Four KPI tiles; a Gender filter and risk colour legend that update every chart at once; a plain-language headline on each chart; a captioned story with a closing "What it all means" takeaway; a web portfolio that embeds the dashboard.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Personal laptop with a multi-core CPU; Intel Arc 130V
Memory	RAM specifications	16 GB RAM
Storage	Disk space for data, models, and logs	Under 1 GB (one Excel file plus Tableau workbook)
Software		
Frameworks	Python frameworks	Flask (for the ePortfolio site); Tableau Public for dashboards and story
Libraries	Additional libraries	HTML, CSS and JavaScript (Tableau Public embed code)
Development Environment	IDE, version control	Tableau Public, VS Code, Git
Data		
Data	Source, size, format	Personalized_Psychological_Intervention_Dataset.xlsx; 1,000 records, 16 student fields; Excel (.xlsx)